6.14 Mixed Practice

AP Calculus 6.14 · Student sheet

Name _____ Period ____ Date _____

I can…

- recognize the form of an integrand.
- choose an integration technique and explain why.
- carry out the technique I chose.

Vocabulary · fill in the missing word

Form of the integrand — the pattern in an integral that points to a _____

Part 1 · Sort the integrals

Which technique?

Integral	Technique
$\int x(x^2 + 1)^5\,dx$	
$\int x \cos x\,dx$	
$\int dx/(x^2 - 1)$	
$\int \sqrt{(9 - x^2)}\,dx$	
$\int \sin^2 x\,dx$	
$\int (x^2 + 1)/(x + 1)\,dx$	

Part 2 · Notes and examples (fill in each note, then work the example)

First: look for a function and its derivative, then try _____.

Example 1

Find $\int x(x^2 + 1)^5\,dx$.

Products like x times e^x, a trig function, or ln x: integration by _____.

Example 2

Find $\int xe^x\,dx$.

Rational functions: long division, then partial _____.

Example 3

Find $\int dx/(x^2 - 1)$.

Square roots $\sqrt{(a^2 \pm x^2)}$ or $\sqrt{(x^2 - a^2)}$: trigonometric _____.

Example 4

Find $\int dx/\sqrt{9 - x^2}$.

Example 5

Find $\int \sin^2 x \, dx$.

Part 3 · Partner practice

Partner 1

Find $\int \frac{e^x}{1 + e^x} \, dx$.

Partner 2

(AP® Practice) Which technique should be used first to find $\int x^2 \ln x \, dx$?

(A) integration by parts (B) u-substitution with $u = x^2$

(C) partial fractions (D) trigonometric substitution